

Practice Exam Two: Version Two

Topics Covered:

- Node Voltage Analysis
- Mesh Current Analysis
- Thévenin and Norton Equivalent Circuits
- Maximum Power Transfer with a Load Resistor
- Equivalent Resistance Calculation including Dependent Sources

Practice Exam Tips:

- Treat this as timed practice, but do not rush. The goal is to identify gaps in understanding.
- After completing the exam, you can grade your work using the provided rubric in the solution. The practice solutions also include suggested discussion questions to try.
- If you make mistakes, revisit the relevant topic and rework the problem before reviewing the solution or attempting the second practice exam.
- Focus on method and clarity, not speed.

Exam Tips:

- Show all work clearly. Correct reasoning with minor algebra errors can still earn partial credit.
- Clearly label currents, voltages, and reference directions.
- Use consistent sign conventions throughout each problem.
- Include units in all final answers.
- The actual exam time limit is 75 minutes. If you follow the suggested pacing from this practice exam, you will finish in 60 minutes.

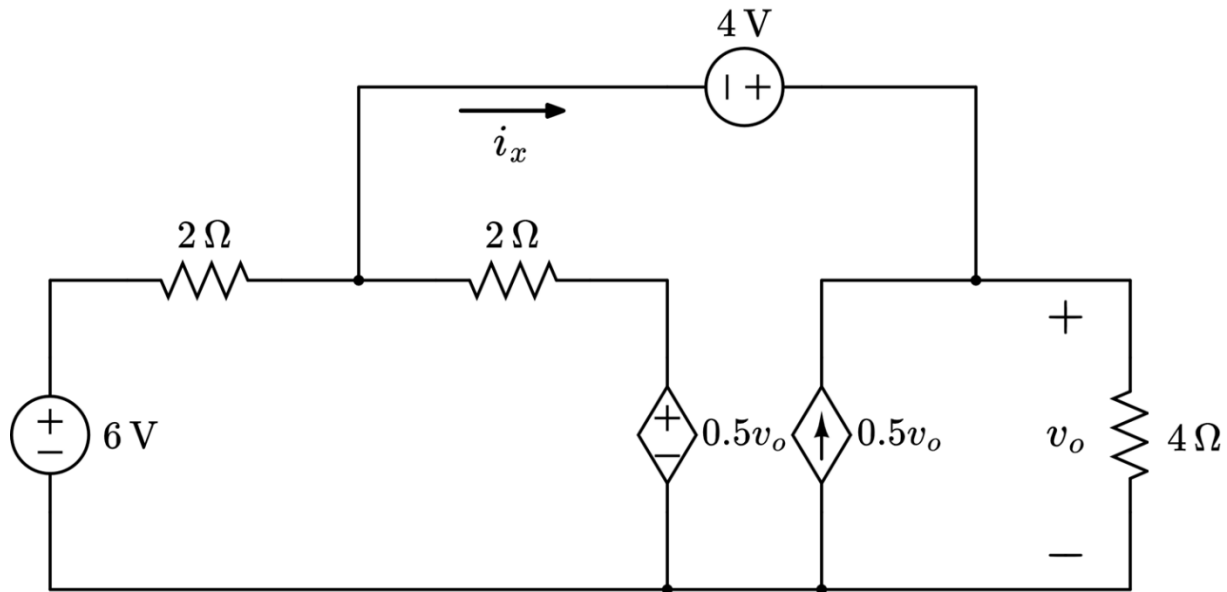
Question One: Node Voltage Analysis

For the circuit shown below, use node voltage analysis, find the following:

- a) v_o
- b) i_x

Using mesh current analysis will result in zero credit earned, regardless of the final answer.

Suggested time limit: 20 minutes



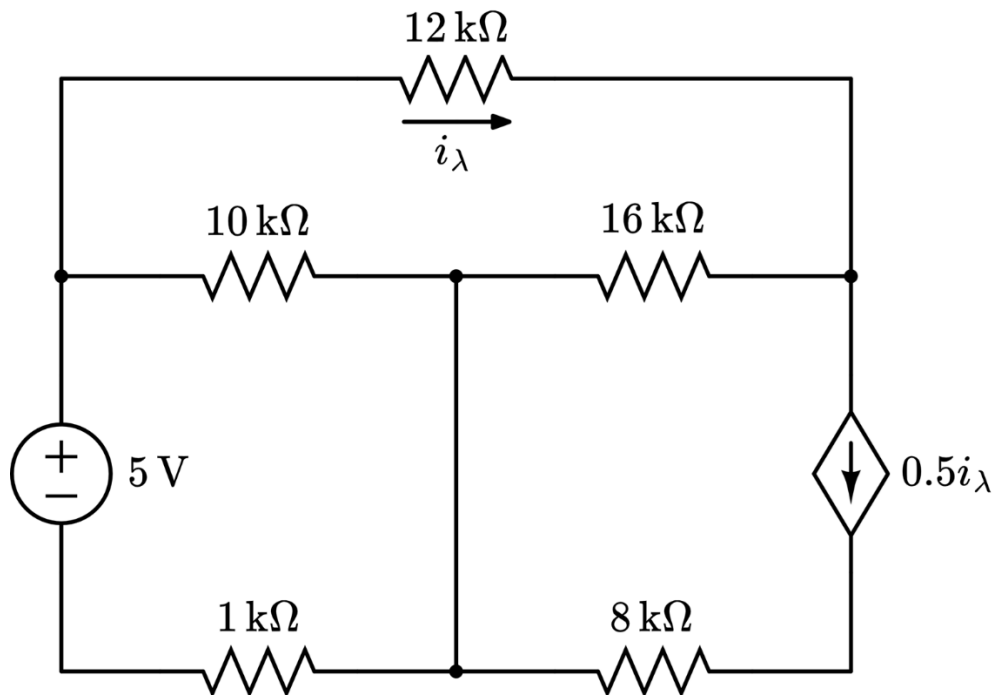
Question Two: Mesh Current Analysis

For the circuit shown below, use mesh current analysis, label and solve for each mesh current.

Finally, calculate the power of the current-dependent current source and state whether the power is absorbed or delivered.

Using node voltage analysis will result in zero credit earned, regardless of the final answer.

Suggested time limit: 20 minutes



Question Three: Thévenin and Norton Equivalent Circuits with Maximum Power Transfer

For the circuit shown below:

- Find the Thévenin equivalent circuit with respect to the open terminal pair ab and clearly illustrate the equivalent circuit with every value labeled.
- If a load is connected to points a and b , what is the maximum power that can be delivered to a load resistor R_L ?
- Suppose that a $22\ \Omega$ resistor is connected to points a and b . Find the voltage, current and power of the $22\ \Omega$ resistor and state whether the resistor is absorbing or delivering power.
Node Voltage, Mesh Current, Current/Voltage Division and KVL/KCL are not needed.

Suggested time limit: 20 minutes

